COVID-19 Sentiment Analysis Case Study Rubric

DS 4002 – Spring 2024 - Lily Thomson
Due: TBD
Submission Format: Upload link to GitHub repository on UVA Canvas Class SIte

Individual Assignment

Why am I doing this? This is your opportunity to apply the skills you have learned as a data scientist student to a real-world scenario. You will use the VADER sentiment analysis tool to evaluate public opinion before and after the initial major COVID-19 shutdown announcements in 2020. This case study will help you understand how to perform sentiment analysis and how to extract meaningful insights and present them in a way that could inform decision-makers during future public health events.

What am I going to do? You will analyze Twitter data from March and April 2020 to compare public sentiment in the U.S. before and after the initial COVID-19 shutdown announcements. You will leverage your technical and conceptual skills to clean the dataset, extract relevant information, and apply the VADER sentiment analysis. You will then use this data to average the average sentiment scores of the tweets on two specific dates- March 16th, 2020, and April 5th, 2020)- and draw conclusions based on the differences. Finally, you will summarize your findings in a concise document that could help guide future decisions.

Deliverables include:

- A Github repository containing:
  - Source code
  - Datasets
  - A written document in PDF format, which includes:
    - A pre-analysis plan
    - An executive summary of findings
    - A reflection paragraph

Tips for success:

- Spend time developing a thoughtful pre-analysis plan.
- Understand how VADER sentiment analysis works before applying it.
- Tailor the executive summary to the intended audience, focusing on key insights and implications.

How will I know I have Succeeded? You will meet expectations if your submission meets the criteria outlined in the rubric below.

Spec Category	Spec Details
Formatting	• One Github Repository (submitted via link on Canvas) o A README.md file (displays automatically) o A LICENSE.md file (use MIT license) o Written portion & Output

<table>
<tr><td></td><td>
Submit as PDF
Data & Source Code
Include all source code and datasets
Must be in an executable Jupyter notebook
References
Include any additional references in README.md file
Use IEEE citation style
</td></tr>
<tr><td>README.md</td><td>
High-level overview of findings and what was completed for the case study (use markdown headers to divide content)
Include the types of software you used for the case study, the platform used (e.g. Windows, Mac, or Linux), and the name of any add-on packages that need to be installed with the software
Include a map of your documentation of the case study in the github site (outline or tree)
</td></tr>
<tr><td>Written portion</td><td>
Include a written portion in a PDF format of the following components:
Before analysis is completed:
One paragraph explaining the motivation for this case study
One paragraph outlining your analysis plan and deliverables
After analysis is completed:
An executive summary document suitable for government/health officials
Include important statistics and graphs to support the conclusions drawn from the case study
State clearly whether there was a statically significant difference in sentiment before and after the shutdown announcements
One paragraph discussing any challenges faced completing the deliverables and how you overcame these. Include key takeaways from the project.
</td></tr>
<tr><td>Source Code</td><td>
The source code should be a well documented Juypter notebook file that can be run to complete the following steps:
Clean the dataset and perform exploratory data analysis (EDA) to visualize initial trends in sentiment data
Orient to “DATA” folder
Use “Corona_NLP_train.csv”
Apply VADER sentiment analysis and filter for only tweets from the United States
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	◦ Save the output as “df_us_with_sentiment_csv” that includes cleaned data set with the new VADER score column • Compare sentiment scores and run statistical tests ◦ Use “df_us_with_sentiment_csv” to compute the average sentiment scores for March 16, 2020 and April 5, 2020 ◦ Generate output figures comparing sentiment trends ◦ Perform an independent t-test to assess statistical significance
DATA Folder	Must include: • Original dataset: “Corona_NLP_train.csv” • Cleaned dataset: “df_us_with_sentiment_csv”

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from [Streifer & Palmer (2020)](#).